Name & Surname **Polimi ID**

Exercise 1

A. What kind of problems may create a parallel scatter pattern? Please discuss and provide examples. (1)

B. Could you please explain what PCAM design methodology is? (1)

C. Under what conditions can the following code be parallelized? Is there a way to write the code for better efficiency if these conditions are met? (2)

```
double total_sum = 0.0;

for (i = 0; i < 100; i++)
   for (j = 2; j < 100; j += 2) {
      a[i][j] = f(a[i][j]) + b[i][j-2];
      total_sum += a[i][j];
   }

for (i = 0; i < 100; i++)
   for (j = 3; j < 100; j += 2) {
      c[i][j] = f(a[i][j]) + b[i][j-2];
      total_sum += c[i][j];
   }
```

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Exercise 2

A. Please describe the parallel gather pattern, including assumptions, complexity, and examples. (2)

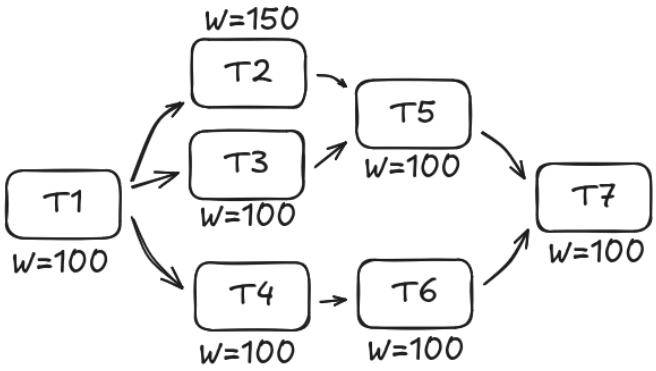
B. Please describe how the Pack parallel pattern works. How is it implemented (pseudo code required)? (2)

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Exercise 3

A. Provide an example of how OpenMP can hinge on both the parallelism across different CPU cores and data parallelism within each one? (1)

B. Given the structure of the following parallel algorithm:



what is the average number of processes that are not idle? Is it better to run it with 2 or 3 threads? (1)

C. **When** should you use `#pragma omp cancel <construct>`? **Why do** you also need `#pragma omp cancellation point?` (2)

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Exercise 4

A. Why is it important to ensure to write standard-compliant MPI code, besides checking if it works on the target testbed? (1) **This question can be skipped if you got 1 or 3 points in the second challenge.**

B. What is an MPI communicator? Provide an example where it makes sense to create multiple communicators. (Generic or vague examples do not count) (1)

C. Provide an example where MPI non-blocking functions do not solve a deadlock. When you want to overlap communication with computation, what are the alternatives that you have, including both MPI and/or OpenMP? (2) **This question can be skipped if you got 2 or 3 points in the second challenge.**